

Chapter 15 GAUSS'S LAW

- 15.1 Charge and Electric Flux
- 15.2 Calculating Electric Flux
- 15.3 Gauss' s Law
- 15.4 Applications of Gauss' s Law
- 15.5 Charges on Conductors

Goals for chapter 15

- How you can determine the amount of charge within a closed surface by examining the electric field on the surface.
- What is meant by electric flux, and how to calculate it.
- How Gauss's law relates the electric flux through a closed surface to the charge enclosed by the surface.
- How to use Gauss's law to calculate the electric field due to a symmetric charge distribution.
- Where the charge is located on a charged conductor.

Introduction



- This child acquires an electric charge by touching the charged metal sphere. The charged hairs on the child's head repel and stand out. If the child stands inside a large, charged metal sphere, will her hair stand on end?

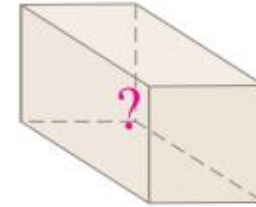
15.1 Charge and Electric Flux

- Given a charge distribution, what is the electric field produced by that distribution at a point ?

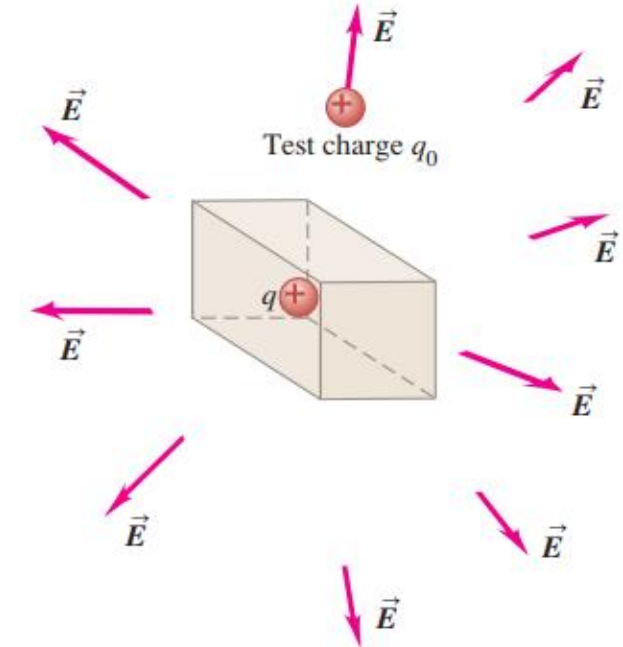
$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \quad (\text{electric field of a point charge})$$

1 How can you measure the charge inside a box without opening it?

(a) A box containing an unknown amount of charge



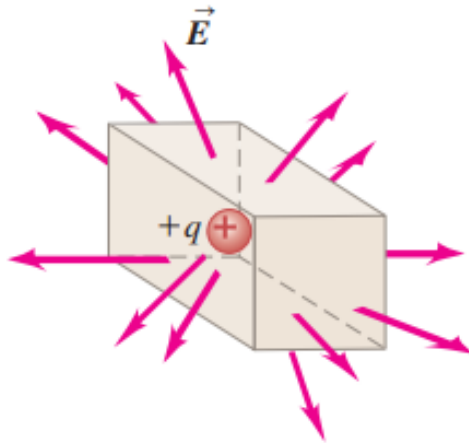
(b) Using a test charge outside the box to probe the amount of charge inside the box



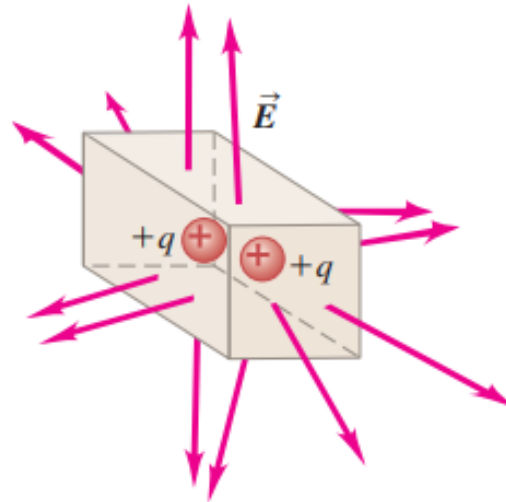
Electric Flux and Enclosed Charge

2 The electric field on the surface of boxes containing (a) a single positive point charge, (b) two positive point charges, (c) a single negative point charge, or (d) two negative point charges.

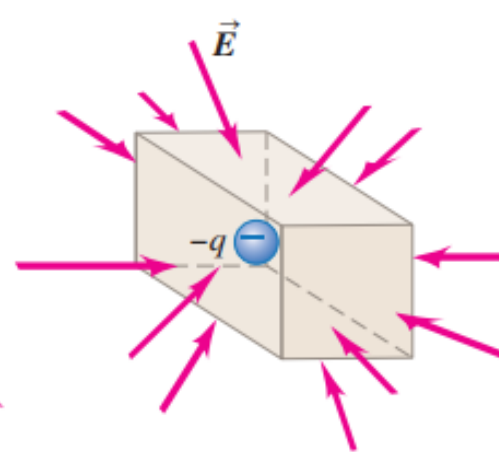
(a) Positive charge inside box,
outward flux



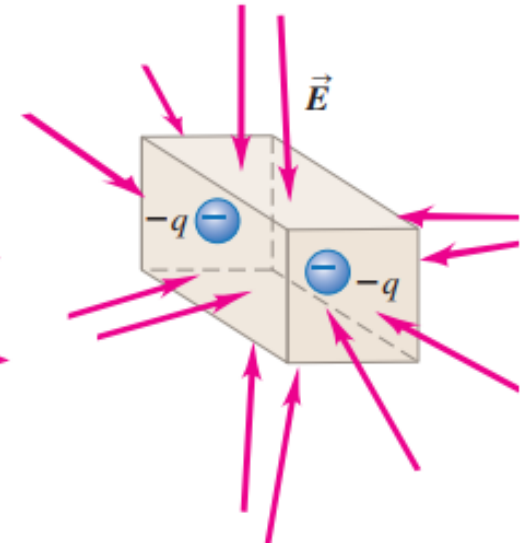
(b) Positive charges inside box,
outward flux



(c) Negative charge inside box,
inward flux



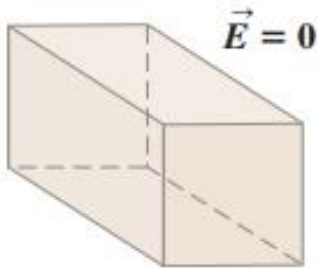
(d) Negative charges inside box,
inward flux



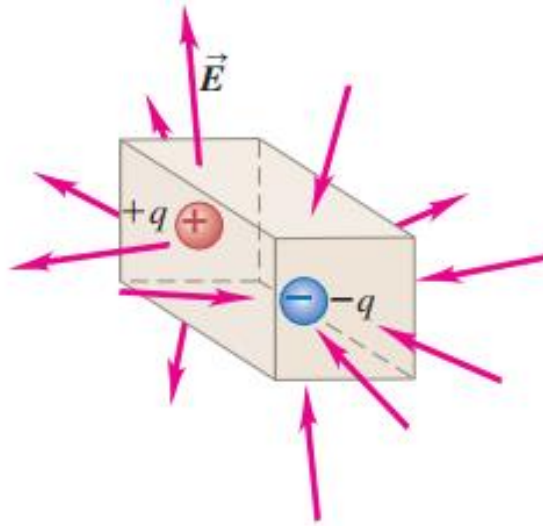
Electric Flux and Enclosed Charge

3 Three cases in which there is zero *net* charge inside a box and no net electric flux through the surface of the box. (a) An empty box with $\vec{E} = \mathbf{0}$. (b) A box containing one positive and one equal-magnitude negative point charge. (c) An empty box immersed in a uniform electric field.

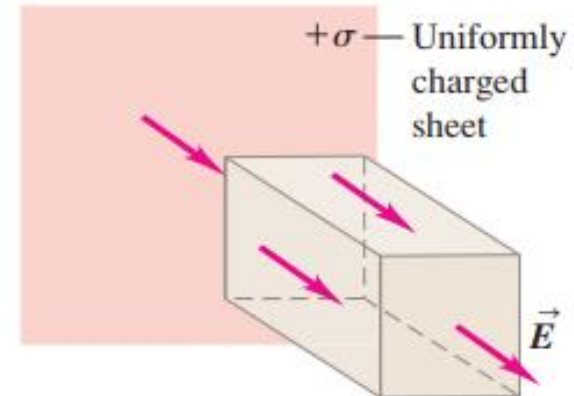
(a) No charge inside box,
zero flux



(b) Zero *net* charge inside box,
inward flux cancels outward flux.



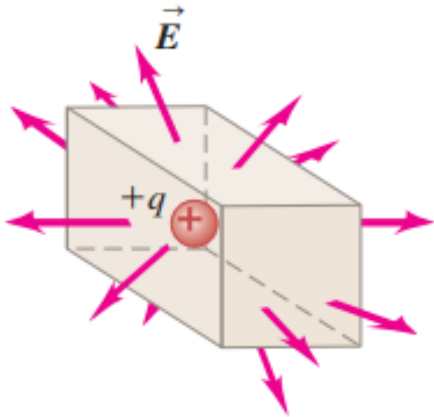
(c) No charge inside box,
inward flux cancels outward flux.



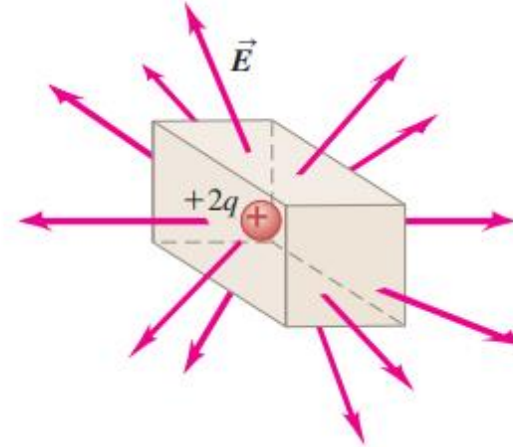
Electric Flux and Enclosed Charge

4 (a) A box enclosing a positive point charge $+q$. (b) Doubling the charge causes the magnitude of $\vec{E}$ to double, and it doubles the electric flux through the surface. (c) If the charge stays the same but the dimensions of the box are doubled, the flux stays the same. The magnitude of $\vec{E}$ on the surface decreases by a factor of $\frac{1}{4}$, but the area through which $\vec{E}$ “flows” increases by a factor of 4.

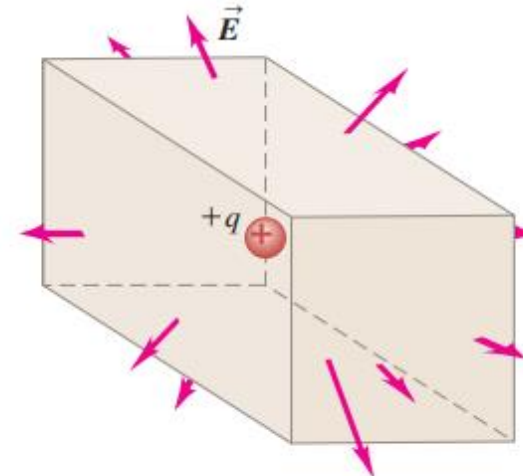
(a) A box containing a charge



(b) Doubling the enclosed charge doubles the flux.



(c) Doubling the box dimensions does not change the flux.



Electric Flux and Enclosed Charge

- To summarize, for the special cases of a closed surface in the shape of a rectangular box and charge distributions made up of point charges or infinite charged sheets, we have found:
- 1. Whether there is a net outward or inward electric flux through a closed surface depends on the sign of the enclosed charge.
- 2. Charges outside the surface do not give a net electric flux through the surface.
- 3. The net electric flux is directly proportional to the net amount of charge enclosed within the surface but is otherwise independent of the size of the closed surface.

15.2 Calculating Electric Flux

- When the area is perpendicular to the flow velocity
- the volume flow rate:

$$\frac{dV}{dt} = vA$$

- When the rectangle is tilted at an angle (Fig. 5b) so that its face is not perpendicular to the flow velocity

$$\frac{dV}{dt} = vA \cos \phi$$

$$\frac{dV}{dt} = v_{\perp} A$$

$$\frac{dV}{dt} = \vec{v} \cdot \vec{A}$$

Flux of a Uniform Electric Field

We define the electric flux through this area to be the product of the field magnitude E and the area A

$$\Phi_E = EA$$

$$\Phi_E = EA \cos \phi \quad (\text{electric flux for uniform } \vec{E}, \text{ flat surface})$$

$$\Phi_E = E_{\perp} A \quad (\text{electric flux for uniform } \vec{E}, \text{ flat surface})$$

$$\Phi_E = \vec{E} \cdot \vec{A} \quad (\text{electric flux for uniform } \vec{E}, \text{ flat surface})$$

We can represent the direction of a vector area $\vec{A}$ by using a *unit vector* $\hat{n}$ perpendicular to the area; $\hat{n}$ stands for “normal.” Then

$$\vec{A} = A\hat{n} \quad [4]$$

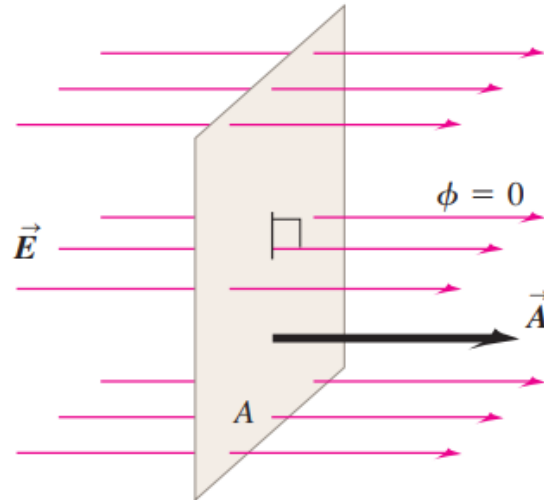
The SI unit for electric flux is $1 \text{ N}\cdot\text{m}^2/\text{C}$, Note that if the area is edge-on to the field, $\vec{E}$ and $\vec{A}$ are perpendicular and the flux is zero.

Flux of a Uniform Electric Field

6 A flat surface in a uniform electric field. The electric flux Φ_E through the surface equals the scalar product of the electric field $\vec{E}$ and the area vector $\vec{A}$.

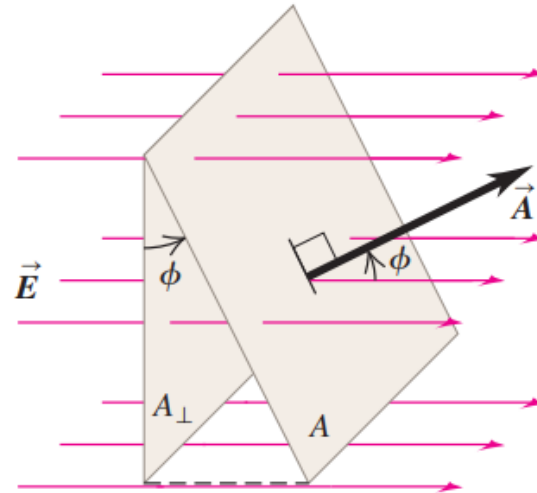
(a) Surface is face-on to electric field:

- $\vec{E}$ and $\vec{A}$ are parallel (the angle between $\vec{E}$ and $\vec{A}$ is $\phi = 0$).
- The flux $\Phi_E = \vec{E} \cdot \vec{A} = EA$.



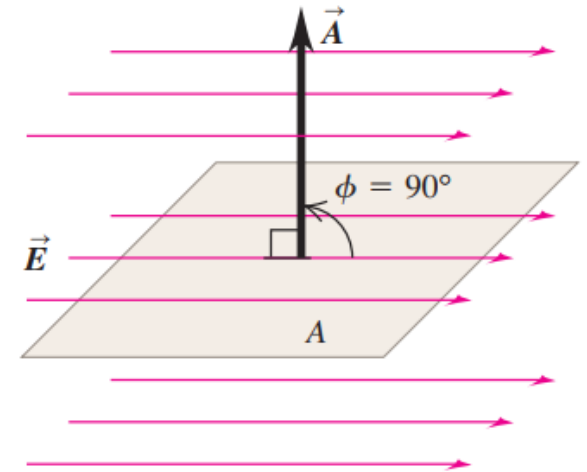
(b) Surface is tilted from a face-on orientation by an angle ϕ :

- The angle between $\vec{E}$ and $\vec{A}$ is ϕ .
- The flux $\Phi_E = \vec{E} \cdot \vec{A} = EA \cos \phi$.



(c) Surface is edge-on to electric field:

- $\vec{E}$ and $\vec{A}$ are perpendicular (the angle between $\vec{E}$ and $\vec{A}$ is $\phi = 90^\circ$).
- The flux $\Phi_E = \vec{E} \cdot \vec{A} = EA \cos 90^\circ = 0$.



Flux of a Nonuniform Electric Field

We calculate the electric flux through each element and integrate the results to obtain the total flux:

$$\Phi_E = \int E \cos \phi \, dA = \int E_{\perp} \, dA = \int \vec{E} \cdot d\vec{A} \quad \text{(general definition of electric flux)}$$

Example 15.1

Electric flux through a disk

A disk of radius 0.10 m is oriented with its normal unit vector $\hat{n}$ at 30° to a uniform electric field $\vec{E}$ of magnitude $2.0 \times 10^3 \text{ N/C}$ (Fig. 7). (Since this isn't a closed surface, it has no "inside" or "outside." That's why we have to specify the direction of $\hat{n}$ in the figure.) (a) What is the electric flux through the disk? (b) What is the flux through the disk if it is turned so that $\hat{n}$ is perpendicular to $\vec{E}$? (c) What is the flux through the disk if $\hat{n}$ is parallel to $\vec{E}$?

SOLUTION

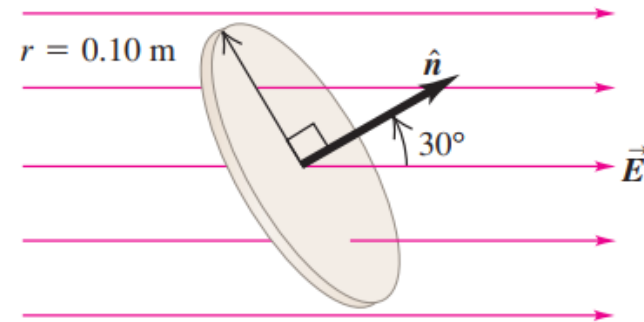
IDENTIFY and SET UP: This problem is about a flat surface in a uniform electric field, so we can apply the ideas of this section. We calculate the electric flux using Eq. (1).

EXECUTE: (a) The area is $A = \pi(0.10 \text{ m})^2 = 0.0314 \text{ m}^2$ and the angle between $\vec{E}$ and $\vec{A} = A\hat{n}$ is $\phi = 30^\circ$, so from Eq. (1),

$$\begin{aligned}\Phi_E &= EA \cos \phi = (2.0 \times 10^3 \text{ N/C})(0.0314 \text{ m}^2)(\cos 30^\circ) \\ &= 54 \text{ N} \cdot \text{m}^2/\text{C}\end{aligned}$$

(b) The normal to the disk is now perpendicular to $\vec{E}$, so $\phi = 90^\circ$, $\cos \phi = 0$, and $\Phi_E = 0$.

7 The electric flux Φ_E through a disk depends on the angle between its normal $\hat{n}$ and the electric field $\vec{E}$.



(c) The normal to the disk is parallel to $\vec{E}$, so $\phi = 0$ and $\cos \phi = 1$:

$$\begin{aligned}\Phi_E &= EA \cos \phi = (2.0 \times 10^3 \text{ N/C})(0.0314 \text{ m}^2)(1) \\ &= 63 \text{ N} \cdot \text{m}^2/\text{C}\end{aligned}$$

EVALUATE: As a check on our results, note that our answer to part (b) is smaller than that to part (a), which is in turn smaller than that to part (c). Is all this as it should be?

Example 15.2

Electric flux through a cube

An imaginary cubical surface of side L is in a region of uniform electric field $\vec{E}$. Find the electric flux through each face of the cube and the total flux through the cube when (a) it is oriented with two of its faces perpendicular to $\vec{E}$ (Fig. 8a) and (b) the cube is turned by an angle θ about a vertical axis (Fig. 8b).

SOLUTION

IDENTIFY and SET UP: Since $\vec{E}$ is uniform and each of the six faces of the cube is flat, we find the flux Φ_{Ei} through each face using Eqs. (3) and (4). The total flux through the cube is the sum of the six individual fluxes.

EXECUTE: (a) Figure 8a shows the unit vectors $\hat{n}_1$ through $\hat{n}_6$ for each face; each unit vector points *outward* from the cube's closed surface. The angle between $\vec{E}$ and $\hat{n}_1$ is 180° , the angle between $\vec{E}$

and $\hat{n}_2$ is 0° , and the angle between $\vec{E}$ and each of the other four unit vectors is 90° . Each face of the cube has area L^2 , so the fluxes through the faces are

$$\Phi_{E1} = \vec{E} \cdot \hat{n}_1 A = EL^2 \cos 180^\circ = -EL^2$$

$$\Phi_{E2} = \vec{E} \cdot \hat{n}_2 A = EL^2 \cos 0^\circ = +EL^2$$

$$\Phi_{E3} = \Phi_{E4} = \Phi_{E5} = \Phi_{E6} = EL^2 \cos 90^\circ = 0$$

The flux is negative on face 1, where $\vec{E}$ is directed into the cube, and positive on face 2, where $\vec{E}$ is directed out of the cube. The total flux through the cube is

$$\begin{aligned} \Phi_E &= \Phi_{E1} + \Phi_{E2} + \Phi_{E3} + \Phi_{E4} + \Phi_{E5} + \Phi_{E6} \\ &= -EL^2 + EL^2 + 0 + 0 + 0 + 0 = 0 \end{aligned}$$

(b) The field $\vec{E}$ is directed into faces 1 and 3, so the fluxes through them are negative; $\vec{E}$ is directed out of faces 2 and 4, so the fluxes through them are positive. We find

$$\Phi_{E1} = \vec{E} \cdot \hat{n}_1 A = EL^2 \cos (180^\circ - \theta) = -EL^2 \cos \theta$$

$$\Phi_{E2} = \vec{E} \cdot \hat{n}_2 A = +EL^2 \cos \theta$$

$$\Phi_{E3} = \vec{E} \cdot \hat{n}_3 A = EL^2 \cos (90^\circ + \theta) = -EL^2 \sin \theta$$

$$\Phi_{E4} = \vec{E} \cdot \hat{n}_4 A = EL^2 \cos (90^\circ - \theta) = +EL^2 \sin \theta$$

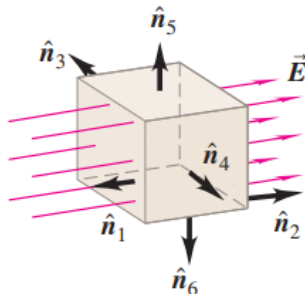
$$\Phi_{E5} = \Phi_{E6} = EL^2 \cos 90^\circ = 0$$

The total flux $\Phi_E = \Phi_{E1} + \Phi_{E2} + \Phi_{E3} + \Phi_{E4} + \Phi_{E5} + \Phi_{E6}$ through the surface of the cube is again zero.

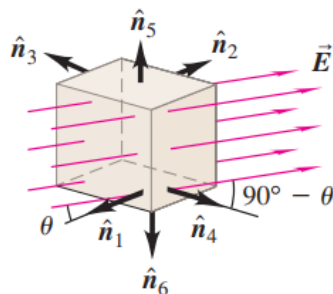
EVALUATE: We came to the same conclusion in our discussion of Fig. 3c: There is zero net flux of a uniform electric field through a closed surface that contains no electric charge.

8 Electric flux of a uniform field $\vec{E}$ through a cubical box of side L in two orientations.

(a)



(b)



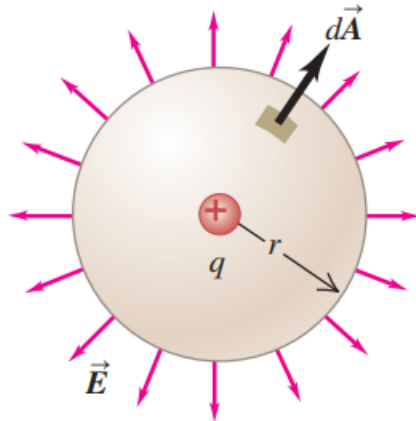
Example 15.3

A point charge $q = +3.0 \mu\text{C}$ is surrounded by an imaginary sphere of radius $r = 0.20 \text{ m}$ centered on the charge (Fig. 9). Find the resulting electric flux through the sphere.

SOLUTION

IDENTIFY and SET UP: The surface is not flat and the electric field is not uniform, so to calculate the electric flux (our target variable)

9 Electric flux through a sphere centered on a point charge.



EXECUTE: We must evaluate the integral of Eq. (5), $\Phi_E = \int E \, dA$. At any point on the sphere of radius r the electric field has the same magnitude $E = q/4\pi\epsilon_0 r^2$. Hence E can be taken outside the integral, which becomes $\Phi_E = E \int dA = EA$, where A is the area

Electric flux through a sphere

we must use the general definition, Eq. (5). We use Eq. (5) to calculate the electric flux (our target variable). Because the sphere is centered on the point charge, at any point on the spherical surface, $\vec{E}$ is directed out of the sphere perpendicular to the surface. The positive direction for both $\hat{n}$ and $E_\perp$ is outward, so $E_\perp = E$ and the flux through a surface element dA is $\vec{E} \cdot d\vec{A} = E \, dA$. This greatly simplifies the integral in Eq. (5).

of the spherical surface: $A = 4\pi r^2$. Hence the total flux through the sphere is

$$\begin{aligned}\Phi_E &= EA = \frac{q}{4\pi\epsilon_0 r^2} 4\pi r^2 = \frac{q}{\epsilon_0} \\ &= \frac{3.0 \times 10^{-6} \text{ C}}{8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2} = 3.4 \times 10^5 \text{ N} \cdot \text{m}^2/\text{C}\end{aligned}$$

EVALUATE: The radius r of the sphere cancels out of the result for Φ_E . We would have obtained the same flux with a sphere of radius 2.0 m or 200 m. We came to essentially the same conclusion in our discussion of Fig. 4 in Section 1, where we considered rectangular closed surfaces of two different sizes enclosing a point charge. There we found that the flux of $\vec{E}$ was independent of the size of the surface; the same result holds true for a spherical surface. Indeed, the flux through *any* surface enclosing a single point charge is independent of the shape or size of the surface, as we'll soon see.

15.3 Gauss's Law

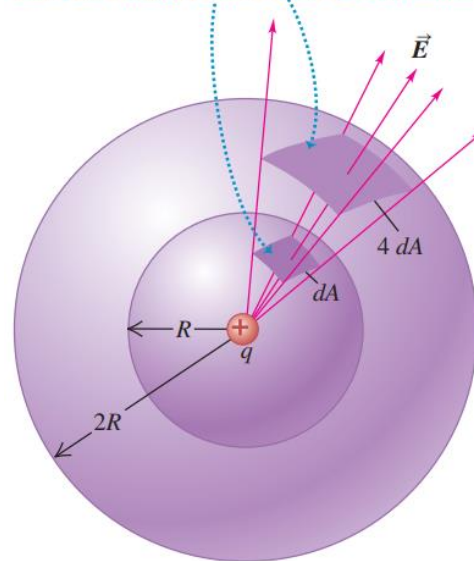
Point Charge Inside a Spherical Surface

- Gauss's law states that the total electric flux through any closed surface (a surface enclosing a definite volume) is proportional to the total (net) electric charge inside the surface.

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2}$$

$$\Phi_E = EA = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2} (4\pi R^2) = \frac{q}{\epsilon_0}$$

The same number of field lines and the same flux pass through both of these area elements.



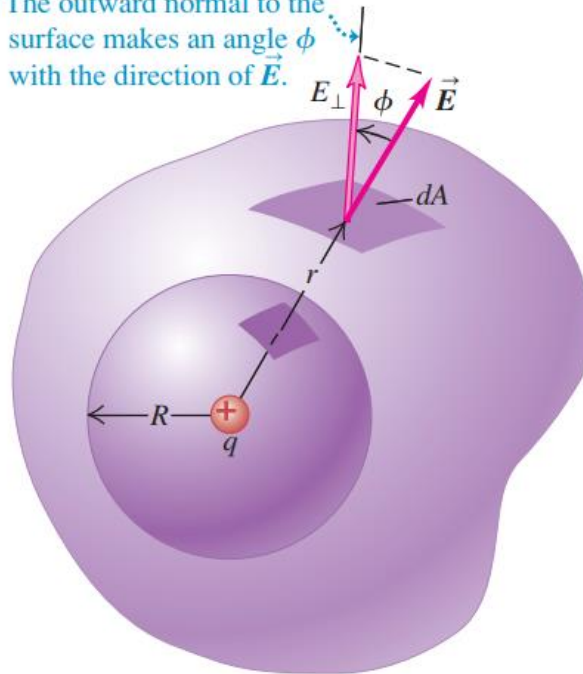
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Point Charge Inside a Nonspherical Surface

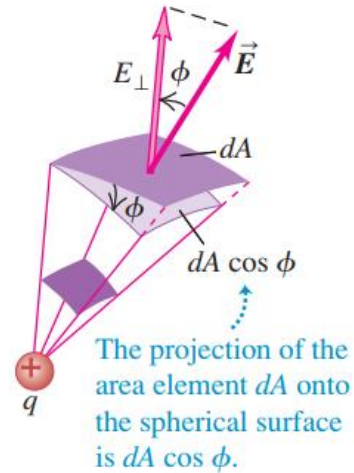
- This projection technique shows us how to extend this discussion to nonspherical surfaces.

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0}$$

(a) The outward normal to the surface makes an angle ϕ with the direction of $\vec{E}$.



(b)

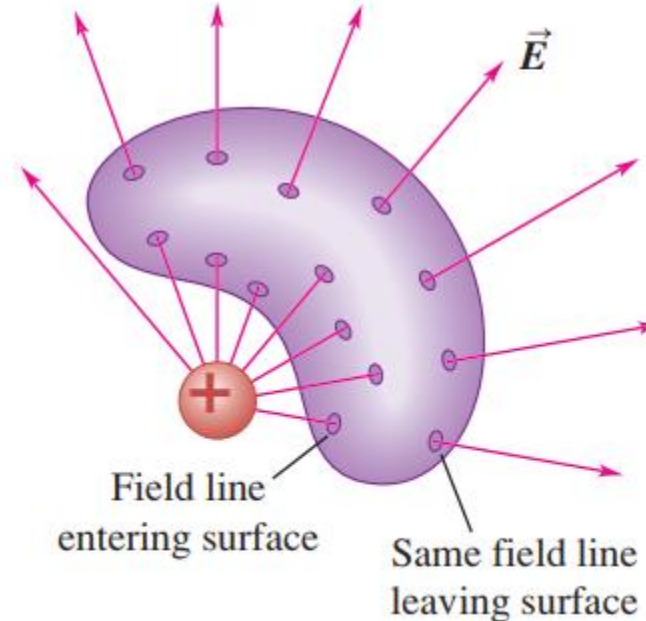


Point Charge Inside a Nonspherical Surface

- For a closed surface enclosing no charge

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = 0$$

13 A point charge *outside* a closed surface that encloses no charge. If an electric field line from the external charge enters the surface at one point, it must leave at another.



General Form of Gauss's Law

we obtain the general statement of Gauss's law:

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0} \quad (\text{Gauss's law})$$

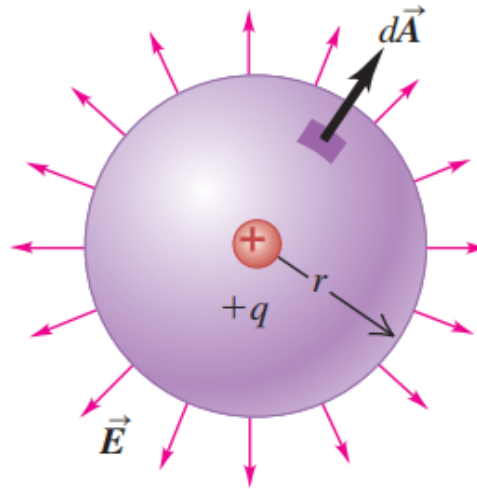
The total electric flux through a closed surface is equal to the total (net) electric charge inside the surface, divided by ϵ_0 .

$$\Phi_E = \oint E \cos \phi \, dA = \oint E_{\perp} \, dA = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0} \quad (\text{various forms of Gauss's law})$$

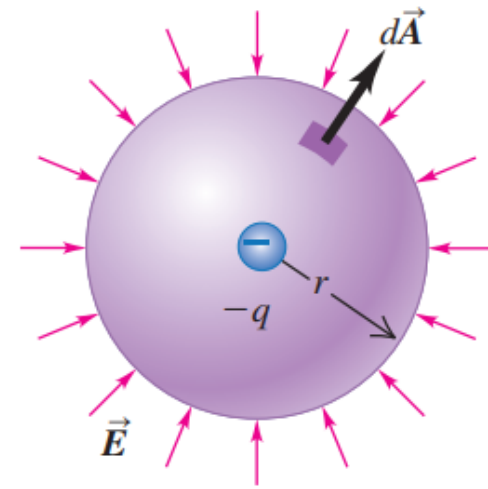
General Form of Gauss's Law

14 Spherical Gaussian surfaces around
(a) a positive point charge and
(b) a negative point charge.

(a) Gaussian surface around positive charge:
positive (outward) flux



(b) Gaussian surface around negative charge:
negative (inward) flux



$$\Phi_E = \oint E_{\perp} dA = \oint \left(\frac{q}{4\pi\epsilon_0 r^2} \right) dA = \frac{q}{4\pi\epsilon_0 r^2} \oint dA = \frac{q}{4\pi\epsilon_0 r^2} 4\pi r^2 = \frac{q}{\epsilon_0}$$

$$\Phi_E = \oint E_{\perp} dA = \oint \left(\frac{-q}{4\pi\epsilon_0 r^2} \right) dA = \frac{-q}{4\pi\epsilon_0 r^2} \oint dA = \frac{-q}{4\pi\epsilon_0 r^2} 4\pi r^2 = \frac{-q}{\epsilon_0}$$

Example 15.4

charge

Electric flux and enclosed

Figure 15 shows the field produced by two point charges $+q$ and $-q$ (an electric dipole). Find the electric flux through each of the closed surfaces A , B , C , and D .

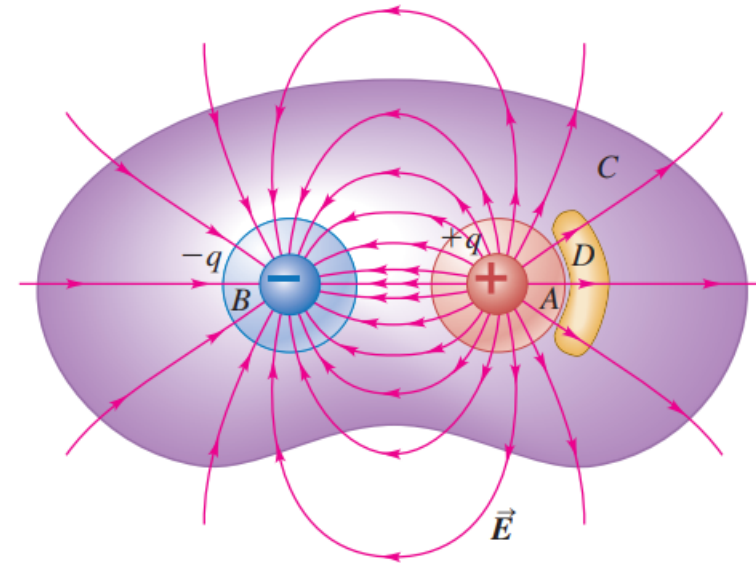
Fig. 15, surface A (shown in red) encloses the positive charge, so $Q_{\text{encl}} = +q$; surface B (in blue) encloses the negative charge, so $Q_{\text{encl}} = -q$; surface C (in purple) encloses *both* charges, so $Q_{\text{encl}} = +q + (-q) = 0$; and surface D (in yellow) encloses no charges, so $Q_{\text{encl}} = 0$. Hence, without having to do any integration, we have $\Phi_{EA} = +q/\epsilon_0$, $\Phi_{EB} = -q/\epsilon_0$, and $\Phi_{EC} = \Phi_{ED} = 0$. These results depend only on the charges enclosed within each Gaussian surface, not on the precise shapes of the surfaces.

We can draw similar conclusions by examining the electric field lines. All the field lines that cross surface A are directed out of the surface, so the flux through A must be positive. Similarly, the flux through B must be negative since all of the field lines that cross that surface point inward. For both surface C and surface D , there are as many field lines pointing into the surface as there are field lines pointing outward, so the flux through each of these surfaces is zero.

SOLUTION

Gauss's law, Eq. (8), says that the total electric flux through a closed surface is equal to the total enclosed charge divided by ϵ_0 . In

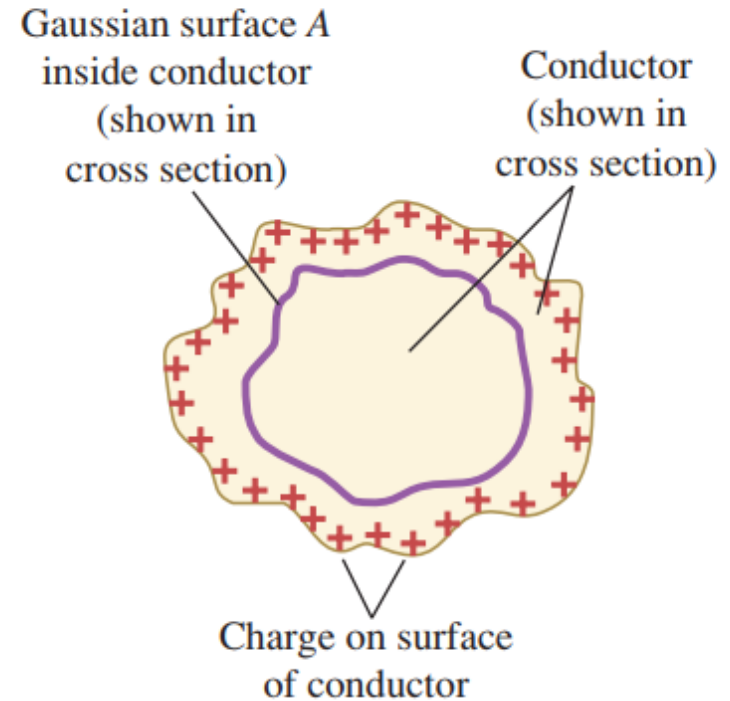
15 The net number of field lines leaving a closed surface is proportional to the total charge enclosed by that surface.



15.4 Applications of Gauss's Law

- These calculations are aided by the following remarkable fact: When excess charge is placed on a solid conductor and is at rest, it resides entirely on the surface, not in the interior of the material. (By excess we mean charges other than the ions and free electrons that make up the neutral conductor.)

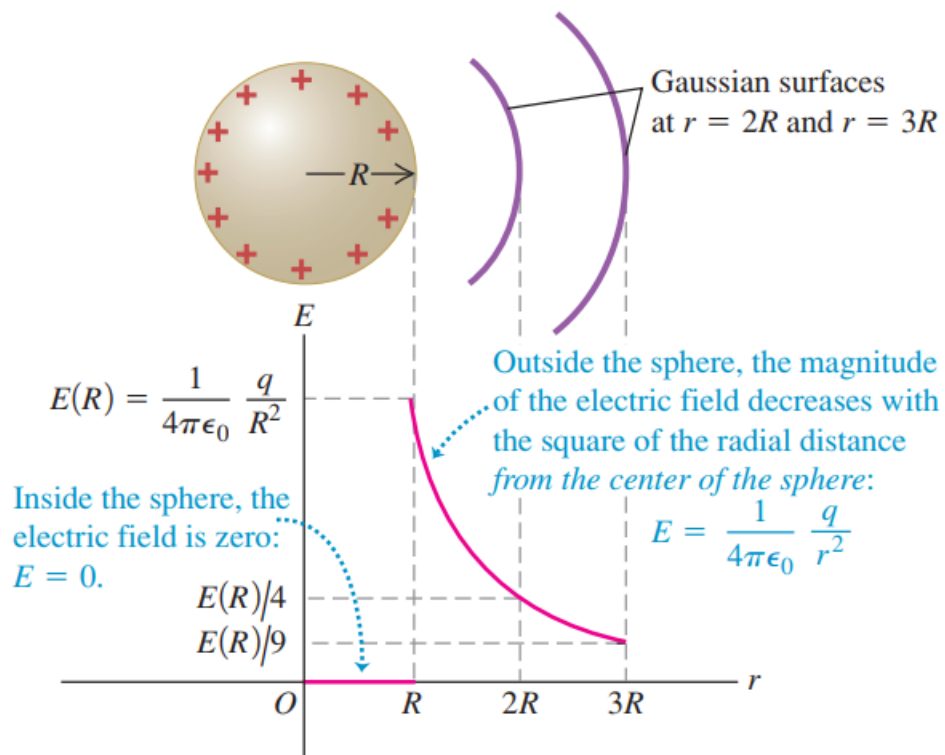
17 Under electrostatic conditions (charges not in motion), any excess charge on a solid conductor resides entirely on the conductor's surface.



Example 15.5 Field of a charged conducting sphere

We place a total positive charge q on a solid conducting sphere with radius R (Fig. 18). Find $\vec{E}$ at any point inside or outside the sphere.

18 Calculating the electric field of a conducting sphere with positive charge q . Outside the sphere, the field is the same as if all of the charge were concentrated at the center of the sphere.



SOLUTION

IDENTIFY and SET UP: As we discussed earlier in this section, all of the charge must be on the surface of the sphere. The charge is free to move on the conductor, and there is no preferred position on the surface; the charge is therefore distributed *uniformly* over the surface, and the system is spherically symmetric. To exploit this symmetry, we take as our Gaussian surface a sphere of radius r centered on the conductor. We can calculate the field inside or outside the conductor by taking $r < R$ or $r > R$, respectively. In either case, the point at which we want to calculate $\vec{E}$ lies on the Gaussian surface.

EXECUTE: The spherical symmetry means that the direction of the electric field must be radial; that's because there is no preferred direction parallel to the surface, so $\vec{E}$ can have no component parallel to the surface. There is also no preferred orientation of the sphere, so the field magnitude E can depend only on the distance r from the center and must have the same value at all points on the Gaussian surface.

For $r > R$ the entire conductor is within the Gaussian surface, so the enclosed charge is q . The area of the Gaussian surface is $4\pi r^2$, and $\vec{E}$ is uniform over the surface and perpendicular to it at each point. The flux integral $\oint \vec{E} \cdot d\vec{A}$ is then just $E(4\pi r^2)$, and Eq. (8) gives

$$E(4\pi r^2) = \frac{q}{\epsilon_0} \quad \text{and}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \quad \text{(outside a charged conducting sphere)}$$

This expression is the same as that for a point charge; outside the charged sphere, its field is the same as though the entire charge were concentrated at its center. Just outside the surface of the sphere, where $r = R$,

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2} \quad \text{(at the surface of a charged conducting sphere)}$$

CAUTION **Flux can be positive or negative** Remember that we have chosen the charge q to be *positive*. If the charge is negative, the electric field is radially *inward* instead of radially outward, and the electric flux through the Gaussian surface is negative. The electric-field magnitudes outside and at the surface of the sphere are given by the same expressions as above, except that q denotes the *magnitude* (absolute value) of the charge. ■

For $r < R$ we again have $E(4\pi r^2) = Q_{\text{encl}}/\epsilon_0$. But now our Gaussian surface (which lies entirely within the conductor)

encloses *no* charge, so $Q_{\text{encl}} = 0$. The electric field inside the conductor is therefore zero.

EVALUATE: We already knew that $\vec{E} = \mathbf{0}$ inside a solid conductor (whether spherical or not) when the charges are at rest. Figure 18 shows E as a function of the distance r from the center of the sphere. Note that in the limit as $R \rightarrow 0$, the sphere becomes a point charge; there is then only an “outside,” and the field is everywhere given by $E = q/4\pi\epsilon_0 r^2$. Thus we have deduced Coulomb’s law from Gauss’s law. (In Section 3 we deduced Gauss’s law from Coulomb’s law; the two laws are equivalent.)

We can also use this method for a conducting spherical *shell* (a spherical conductor with a concentric spherical hole inside) if there is no charge inside the hole. We use a spherical Gaussian surface with radius r less than the radius of the hole. If there *were* a field inside the hole, it would have to be radial and spherically symmetric as before, so $E = Q_{\text{encl}}/4\pi\epsilon_0 r^2$. But now there is no enclosed charge, so $Q_{\text{encl}} = 0$ and $E = 0$ inside the hole.

Can you use this same technique to find the electric field in the region between a charged sphere and a concentric hollow conducting sphere that surrounds it?

Example 15.6 Field of a uniform line charge

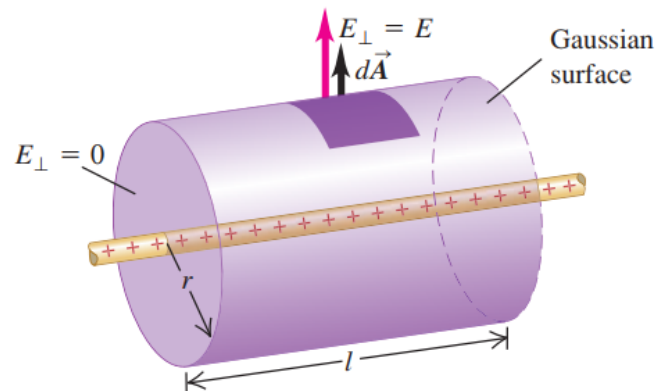
Electric charge is distributed uniformly along an infinitely long, thin wire. The charge per unit length is λ (assumed positive). Find the electric field using Gauss's law.

SOLUTION

IDENTIFY and SET UP: The field $\vec{E}$ of a uniformly charged, infinite wire is radially outward if λ is positive and radially inward if λ is negative, and that the field magnitude E depends only on the radial distance from the wire. This suggests that we use a *cylindrical* Gaussian surface, of radius r and arbitrary length l , coaxial with the wire and with its ends perpendicular to the wire (Fig. 19).

EXECUTE: The flux through the flat ends of our Gaussian surface is zero because the radial electric field is parallel to these ends, and so $\vec{E} \cdot \hat{n} = 0$. On the cylindrical part of our surface we have $\vec{E} \cdot \hat{n} = E_{\perp} = E$ everywhere. (If λ were negative, we would have

19 A coaxial cylindrical Gaussian surface is used to find the electric field outside an infinitely long, charged wire.



$\vec{E} \cdot \hat{n} = E_{\perp} = -E$ everywhere.) The area of the cylindrical surface is $2\pi rl$, so the flux through it—and hence the total flux Φ_E through the Gaussian surface—is $EA = 2\pi rlE$. The total enclosed charge is $Q_{\text{encl}} = \lambda l$, and so from Gauss's law, Eq. (8),

$$\Phi_E = 2\pi rlE = \frac{\lambda l}{\epsilon_0} \quad \text{and}$$

$$E = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{r} \quad (\text{field of an infinite line of charge})$$

If λ is *negative*, $\vec{E}$ is directed radially inward, and in the above expression for E we must interpret λ as the absolute value of the charge per unit length.

EVALUATE: The *entire* charge on the wire contributes to the field at any point, and yet we consider only that part of the charge $Q_{\text{encl}} = \lambda l$ within the Gaussian surface when we apply Gauss's law. There's nothing inconsistent here; it takes the entire charge to give the field the properties that allow us to calculate Φ_E so easily, and Gauss's law always applies to the enclosed charge only. If the wire is short, the symmetry of the infinite wire is lost, and E is not uniform over a coaxial, cylindrical Gaussian surface. Gauss's law then *cannot* be used to find Φ_E ; we must solve the problem the hard way.

We can use the Gaussian surface in Fig. 19 to show that the field outside a long, uniformly charged cylinder is the same as though all the charge were concentrated on a line along its axis (see Problem 42). We can also calculate the electric field in the space between a charged cylinder and a coaxial hollow conducting cylinder surrounding it (see Problem 39).

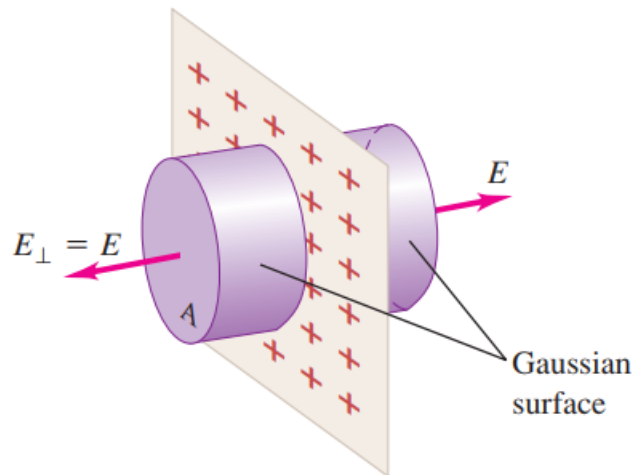
Example 15.7 Field of an infinite plane sheet of charge

Use Gauss's law to find the electric field caused by a thin, flat, infinite sheet with a uniform positive surface charge density σ .

SOLUTION

IDENTIFY and SET UP: The field $\vec{E}$ of a uniformly charged infinite sheet is normal to the sheet, and that its magnitude is independent of the distance from the sheet. To take advantage of these symmetry properties, we use a cylindrical Gaussian surface with ends of area A and with its axis perpendicular to the sheet of charge (Fig. 20).

20 A cylindrical Gaussian surface is used to find the field of an infinite plane sheet of charge.



EXECUTE: The flux through the cylindrical part of our Gaussian surface is zero because $\vec{E} \cdot \hat{n} = 0$ everywhere. The flux through each flat end of the surface is $+EA$ because $\vec{E} \cdot \hat{n} = E_{\perp} = E$ everywhere, so the total flux through both ends—and hence the total flux Φ_E through the Gaussian surface—is $+2EA$. The total enclosed charge is $Q_{\text{encl}} = \sigma A$, and so from Gauss's law,

$$2EA = \frac{\sigma A}{\epsilon_0} \quad \text{and}$$
$$E = \frac{\sigma}{2\epsilon_0} \quad (\text{field of an infinite sheet of charge})$$

If σ is negative, $\vec{E}$ is directed *toward* the sheet, the flux through the Gaussian surface in Fig. 20 is negative, and σ in the expression $E = \sigma/2\epsilon_0$ denotes the magnitude (absolute value) of the charge density.

EVALUATE: Again we see that, given favorable symmetry, we can deduce electric fields using Gauss's law much more easily than using Coulomb's law.

Example 15.8 Field between oppositely charged parallel conducting plates

Two large plane parallel conducting plates are given charges of equal magnitude and opposite sign; the surface charge densities are $+\sigma$ and $-\sigma$. Find the electric field in the region between the plates.

SOLUTION

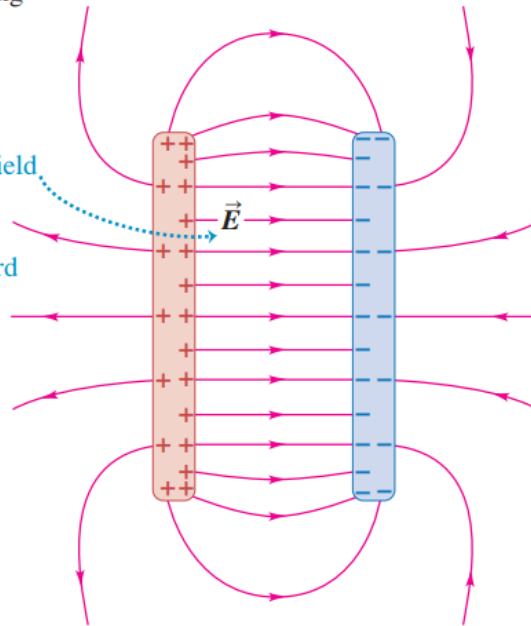
IDENTIFY and SET UP: Figure 21a shows the field. Because opposite charges attract, most of the charge accumulates at the opposing faces of the plates. A small amount of charge resides on the *outer* surfaces of the plates, and there is some spreading or “fringing” of

the field at the edges. But if the plates are very large in comparison to the distance between them, the amount of charge on the outer surfaces is negligibly small, and the fringing can be neglected except near the edges. In this case we can assume that the field is uniform in the interior region between the plates, as in Fig. 21b, and that the charges are distributed uniformly over the opposing surfaces. To exploit this symmetry, we can use the shaded Gaussian surfaces S_1 , S_2 , S_3 , and S_4 . These surfaces are cylinders with flat ends of area A ; one end of each surface lies *within* a plate.

21 Electric field between oppositely charged parallel plates.

(a) Realistic drawing

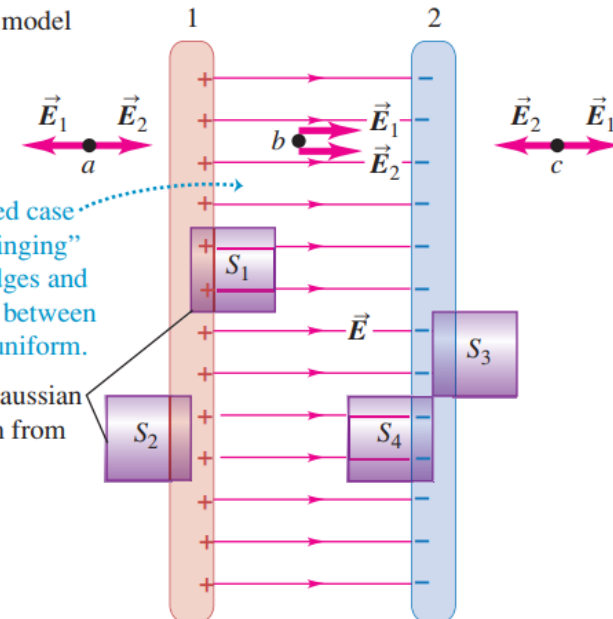
Between the two plates the electric field is nearly uniform, pointing from the positive plate toward the negative one.



(b) Idealized model

In the idealized case we ignore “fringing” at the plate edges and treat the field between the plates as uniform.

Cylindrical Gaussian surfaces (seen from the side)



Example 15.9 Field of a uniformly charged sphere

Positive electric charge Q is distributed uniformly *throughout the volume* of an *insulating* sphere with radius R . Find the magnitude of the electric field at a point P a distance r from the center of the sphere.

SOLUTION

IDENTIFY and SET UP: As in Example 5, the system is spherically symmetric. Hence we can use the conclusions of that example about the direction and magnitude of $\vec{E}$. To make use of the spherical symmetry, we choose as our Gaussian surface a sphere with radius r , concentric with the charge distribution.

EXECUTE: From symmetry, the direction of $\vec{E}$ is radial at every point on the Gaussian surface, so $E_{\perp} = E$ and the field magnitude E is the same at every point on the surface. Hence the total electric flux through the Gaussian surface is the product of E and the total area of the surface $A = 4\pi r^2$ —that is, $\Phi_E = 4\pi r^2 E$.

The amount of charge enclosed within the Gaussian surface depends on r . To find E *inside* the sphere, we choose $r < R$. The volume charge density ρ is the charge Q divided by the volume of the entire charged sphere of radius R :

$$\rho = \frac{Q}{4\pi R^3/3}$$

The volume V_{encl} enclosed by the Gaussian surface is $\frac{4}{3}\pi r^3$, so the total charge Q_{encl} enclosed by that surface is

$$Q_{\text{encl}} = \rho V_{\text{encl}} = \left(\frac{Q}{4\pi R^3/3} \right) \left(\frac{4}{3}\pi r^3 \right) = Q \frac{r^3}{R^3}$$

Then Gauss's law, Eq. (8), becomes

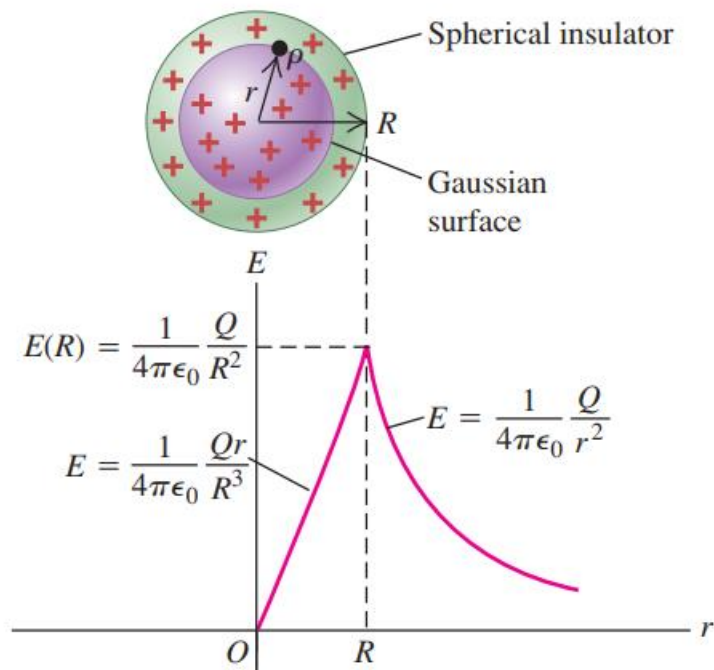
$$4\pi r^2 E = \frac{Q}{\epsilon_0} \frac{r^3}{R^3} \quad \text{or}$$
$$E = \frac{1}{4\pi\epsilon_0} \frac{Qr}{R^3} \quad \text{(field inside a uniformly charged sphere)}$$

The field magnitude is proportional to the distance r of the field point from the center of the sphere (see the graph of E versus r in Fig. 22).

To find E *outside* the sphere, we take $r > R$. This surface encloses the entire charged sphere, so $Q_{\text{encl}} = Q$, and Gauss's law gives

$$4\pi r^2 E = \frac{Q}{\epsilon_0} \quad \text{or}$$
$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \quad \text{(field outside a uniformly charged sphere)}$$

22 The magnitude of the electric field of a uniformly charged insulating sphere. Compare this with the field for a conducting sphere (see Fig. 18).



The field outside *any* spherically symmetric charged body varies as $1/r^2$, as though the entire charge were concentrated at the center. This is graphed in Fig. 22.

If the charge is *negative*, $\vec{E}$ is radially *inward* and in the expressions for E we interpret Q as the absolute value of the charge.

EVALUATE: Notice that if we set $r = R$ in either expression for E , we get the same result $E = Q/4\pi\epsilon_0 R^2$ for the magnitude of the field at the surface of the sphere. This is because the magnitude E is a *continuous* function of r . By contrast, for the charged conducting sphere of Example 5 the electric-field magnitude is *discontinuous* at $r = R$ (it jumps from $E = 0$ just inside the sphere to $E = Q/4\pi\epsilon_0 R^2$ just outside the sphere). In general, the electric field $\vec{E}$ is discontinuous in magnitude, direction, or both wherever there is a *sheet* of charge, such as at the surface of a charged conducting sphere (Example 5), at the surface of an infinite charged sheet (Example 7), or at the surface of a charged conducting plate (Example 8).

The approach used here can be applied to *any* spherically symmetric distribution of charge, even if it is not radially uniform, as it was here. Such charge distributions occur within many atoms and atomic nuclei, so Gauss's law is useful in atomic and nuclear physics.

Example 15.10 Charge on a hollow sphere

A thin-walled, hollow sphere of radius 0.250 m has an unknown charge distributed uniformly over its surface. At a distance of 0.300 m from the center of the sphere, the electric field points radially inward and has magnitude $1.80 \times 10^2 \text{ N/C}$. How much charge is on the sphere?

SOLUTION

IDENTIFY and SET UP: The charge distribution is spherically symmetric. As in Examples 5 and 9, it follows that the electric field is radial everywhere and its magnitude is a function only of the radial distance r from the center of the sphere. We use a spherical Gaussian surface that is concentric with the charge distribution and has radius $r = 0.300 \text{ m}$. Our target variable is $Q_{\text{encl}} = q$.

EXECUTE: The charge distribution is the same as if the charge were on the surface of a 0.250-m-radius conducting sphere. Hence we can borrow the results of Example 5. We note that the electric field

here is directed toward the sphere, so that q must be *negative*. Furthermore, the electric field is directed into the Gaussian surface, so that $E_{\perp} = -E$ and $\Phi_E = \oint E_{\perp} dA = -E(4\pi r^2)$.

By Gauss's law, the flux is equal to the charge q on the sphere (all of which is enclosed by the Gaussian surface) divided by ϵ_0 . Solving for q , we find

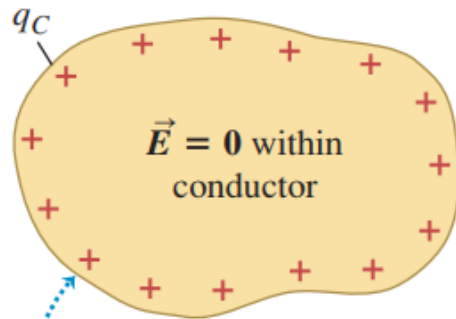
$$\begin{aligned} q &= -E(4\pi\epsilon_0 r^2) = -(1.80 \times 10^2 \text{ N/C})(4\pi) \\ &\quad \times (8.854 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(0.300 \text{ m})^2 \\ &= -1.80 \times 10^{-9} \text{ C} = -1.80 \text{ nC} \end{aligned}$$

EVALUATE: To determine the charge, we had to know the electric field at *all* points on the Gaussian surface so that we could calculate the flux integral. This was possible here because the charge distribution is highly symmetric. If the charge distribution is irregular or lacks symmetry, Gauss's law is not very useful for calculating the charge distribution from the field, or vice versa.

15.5 Charges on Conductors

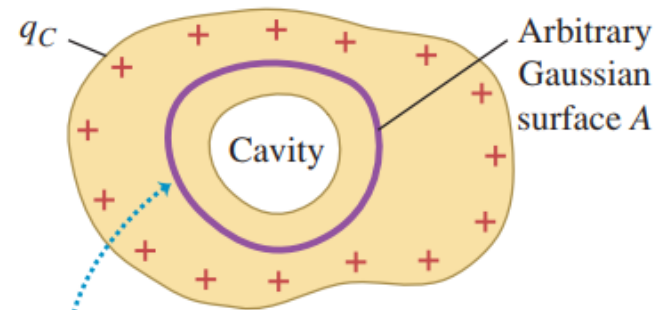
23 Finding the electric field within a charged conductor.

(a) Solid conductor with charge q_C



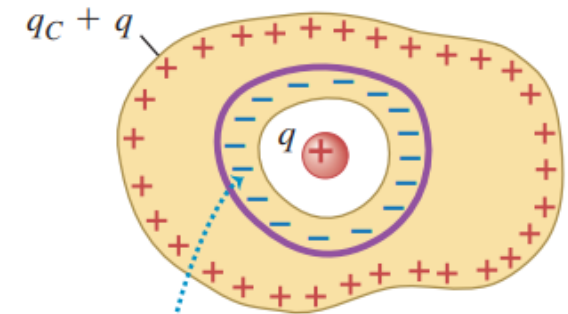
The charge q_C resides entirely on the surface of the conductor. The situation is electrostatic, so $\vec{E} = 0$ within the conductor.

(b) The same conductor with an internal cavity



Because $\vec{E} = 0$ at all points within the conductor, the electric field at all points on the Gaussian surface must be zero.

(c) An isolated charge q placed in the cavity



For $\vec{E}$ to be zero at all points on the Gaussian surface, the surface of the cavity must have a total charge $-q$.

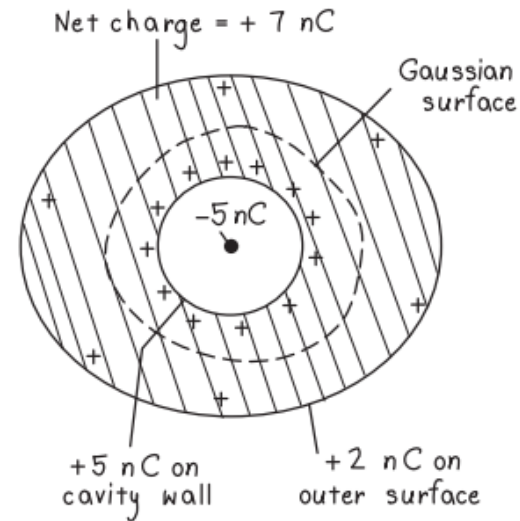
Example 15.11 A conductor with a cavity

A solid conductor with a cavity carries a total charge of $+7 \text{ nC}$. Within the cavity, insulated from the conductor, is a point charge of -5 nC . How much charge is on each surface (inner and outer) of the conductor?

SOLUTION

Figure 24 shows the situation. If the charge in the cavity is $q = -5 \text{ nC}$, the charge on the inner cavity surface must be $-q = -(-5 \text{ nC}) = +5 \text{ nC}$. The conductor carries a *total* charge of $+7 \text{ nC}$, none of which is in the interior of the material. If $+5 \text{ nC}$ is on the inner surface of the cavity, then there must be $(+7 \text{ nC}) - (+5 \text{ nC}) = +2 \text{ nC}$ on the outer surface of the conductor.

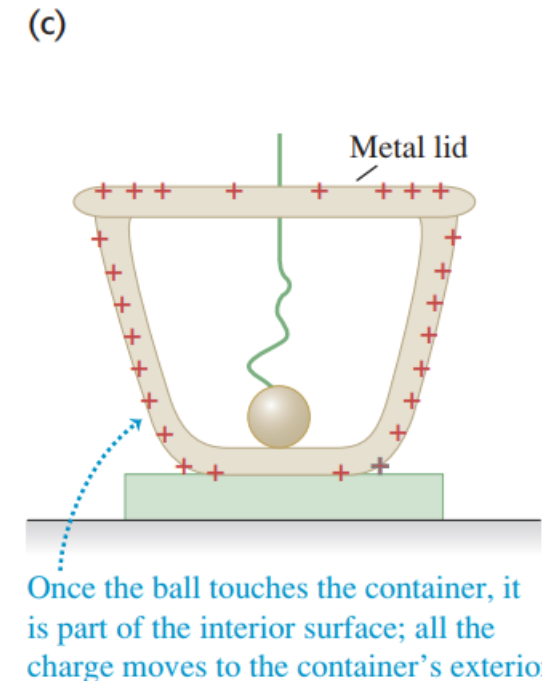
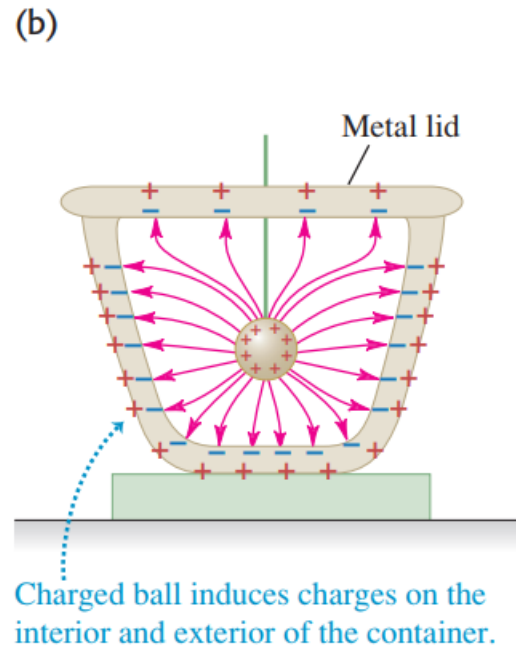
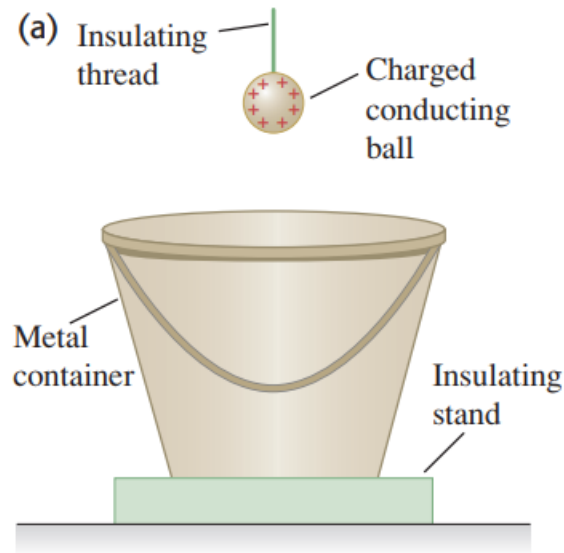
24 Our sketch for this problem. There is zero electric field inside the bulk conductor and hence zero flux through the Gaussian surface shown, so the charge on the cavity wall must be the opposite of the point charge.



Testing Gauss's Law Experimentally

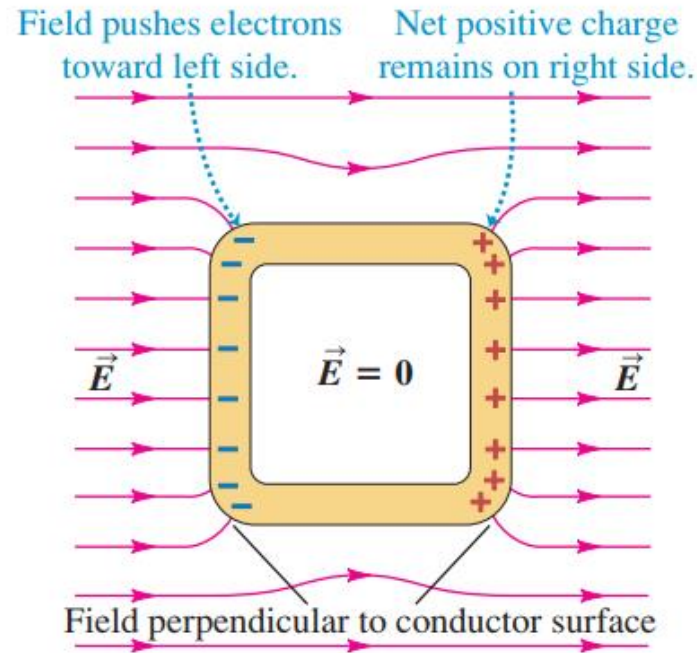
- This experiment was performed in the 19th century by the English scientist Michael Faraday, using a metal icepail with a lid, and it is called Faraday's icepail experiment.

25 (a) A charged conducting ball suspended by an insulating thread outside a conducting container on an insulating stand. (b) The ball is lowered into the container, and the lid is put on. (c) The ball is touched to the inner surface of the container.



27 (a) A conducting box (a Faraday cage) immersed in a uniform electric field. The field of the induced charges on the box combines with the uniform field to give zero total field inside the box. (b) This person is inside a Faraday cage, and so is protected from the powerful electric discharge.

(a)



(b)



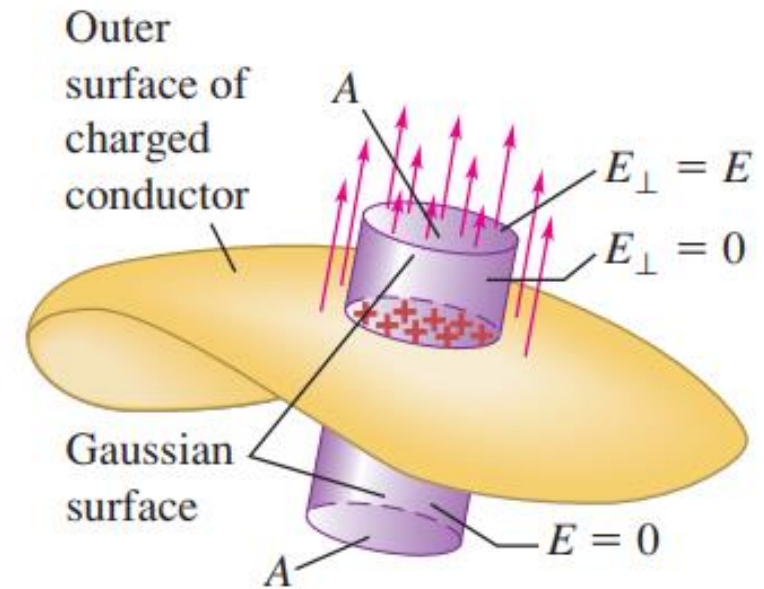
Peter Terren/Tesladowunder.com

Field at the Surface of a Conductor

- To find a relationship between at any point on the surface and the perpendicular component of the electric field at that point, we construct a Gaussian surface in the form of a small cylinder

$$E_{\perp} A = \frac{\sigma A}{\epsilon_0} \quad \text{and} \quad E_{\perp} = \frac{\sigma}{\epsilon_0} \quad (\text{field at the surface of a conductor})$$

28 The field just outside a charged conductor is perpendicular to the surface, and its perpendicular component $E_{\perp}$ is equal to σ/ϵ_0 .



Russ Kinne/Comstock

Example 15.12 Field at the surface of a conducting sphere

Verify Eq. (10) for a conducting sphere with radius R and total charge q .

SOLUTION

In Example 5 (Section 4) we showed that the electric field just outside the surface is

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2}$$

The surface charge density is uniform and equal to q divided by the surface area of the sphere:

$$\sigma = \frac{q}{4\pi R^2}$$

Comparing these two expressions, we see that $E = \sigma/\epsilon_0$, which verifies Eq. (10).

Example 15.13 Electric field of the earth

The earth (a conductor) has a net electric charge. The resulting electric field near the surface has an average value of about 150 N/C, directed toward the center of the earth. (a) What is the corresponding surface charge density? (b) What is the *total* surface charge of the earth?

SOLUTION

IDENTIFY and SET UP: We are given the electric-field magnitude at the surface of the conducting earth. We can calculate the surface charge density σ using Eq. (10). The total charge Q on the earth's surface is then the product of σ and the earth's surface area.

EXECUTE: (a) The direction of the field means that σ is negative (corresponding to $\vec{E}$ being directed *into* the surface, so $E_{\perp}$ is negative). From Eq. (10),

$$\begin{aligned}\sigma &= \epsilon_0 E_{\perp} = (8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(-150 \text{ N/C}) \\ &= -1.33 \times 10^{-9} \text{ C/m}^2 = -1.33 \text{ nC/m}^2\end{aligned}$$

(b) The earth's surface area is $4\pi R_E^2$, where $R_E = 6.38 \times 10^6 \text{ m}$ is the radius of the earth. The total charge Q is the product $4\pi R_E^2 \sigma$, or

$$\begin{aligned}Q &= 4\pi(6.38 \times 10^6 \text{ m})^2(-1.33 \times 10^{-9} \text{ C/m}^2) \\ &= -6.8 \times 10^5 \text{ C} = -680 \text{ kC}\end{aligned}$$

EVALUATE: You can check our result in part (b) using the result of Example 5. Solving for Q , we find

$$\begin{aligned}Q &= 4\pi\epsilon_0 R^2 E_{\perp} \\ &= \frac{1}{9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2}(6.38 \times 10^6 \text{ m})^2(-150 \text{ N/C}) \\ &= -6.8 \times 10^5 \text{ C}\end{aligned}$$

One electron has a charge of $-1.60 \times 10^{-19} \text{ C}$. Hence this much excess negative electric charge corresponds to there being $(-6.8 \times 10^5 \text{ C})/(-1.60 \times 10^{-19} \text{ C}) = 4.2 \times 10^{24}$ excess electrons on the earth, or about 7 moles of excess electrons. This is compensated by an equal *deficiency* of electrons in the earth's upper atmosphere, so the combination of the earth and its atmosphere is electrically neutral.